

CONIC SECTION

- Let S be a fixed point (Focus) and let l be a fixed line (Directrix). Let P be a variable point, then it moves in such a way that $\frac{SP}{PM} = e > 0$, where PM is perpendicular distance from P to the line l , then locus of P is called as conic or conic section.
 - (a) When $e = 1$, conic section is called as **parabola**.
 - (b) When $e < 1$, conic section is called as **ellipse**.
 - (c) When $e > 1$, conic section is called as **hyperbola**.
 - (d) When $e = 0$, conic section is called as **circle**.
 - (e) When $e = \infty$, conic section is called as pair of **straight lines**.
 - (f) **Axis**: The straight line passing through the focus and perpendicular to the directrix of the conic is known as its axis.
 - (g) **Vertex**: A point of intersection of a conic with its axis is called as vertex.
 - (h) **Centre**: The point which bisects every chord of the conic passing through it, is called as centre of the conic.
 - (i) **Focal Chord**: A chord passes through the focus is known as focal chord of the conic.
 - (j) **Latus Rectum**: The focal chord which is perpendicular to its axis is known as latus rectum of the conic.
 - (k) **Double Ordinate**: A chord of the conic which is perpendicular to the axis is called the double ordinate of the conic.
- The general equation of conic is: $S : ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$.

$$\text{Let } \Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = abc + 2fgh - af^2 - bg^2 - ch^2.$$

- (a) If $\Delta \neq 0$, $h^2 = ab$, then $S = 0$ represents a parabola.
- (b) If $\Delta \neq 0$, $h^2 < ab$, then $S = 0$ represents an ellipse.
- (c) If $\Delta \neq 0$, $h^2 > ab$, then $S = 0$ represents a hyperbola.
- (d) If $\Delta \neq 0$, $h^2 > ab$, $a + b = 0$, then $S = 0$ represents a rectangular hyperbola.
- (e) Equation of the tangent to the conic at $P(x_1, y_1)$ is :

$$axx_1 + 2h\left(\frac{xy_1 + x_1y}{2}\right) + byy_1 + g(x + x_1) + f(y + y_1) + c = 0 \quad \text{OR}$$

$$(x - x_1)(ax_1 + hy_1 + g) + (y - y_1)(hx_1 + by_1 + f) = 0.$$

- (f) Equation of the tangent to the conic at $P(x_1, y_1)$ is :

$$\frac{x - x_1}{ax_1 + hy_1 + g} = \frac{y - y_1}{hx_1 + by_1 + f}.$$

- (g) To find the equation of tangent to the conic $S : ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ at $P(x_1, y_1)$ we have to replace:
 - (1) x^2 by xx_1 .
 - (2) y^2 by yy_1 .

$$(3) \quad xy \text{ by } \frac{xy_1 + x_1y}{2}.$$

$$(4) \quad x \text{ by } \frac{x + x_1}{2}.$$

$$(5) \quad y \text{ by } \frac{y + y_1}{2}.$$
